Interfacing LED/LEDS:

1. Flash an LED connected to any port line at the rate of 1 second for 10 times and stop.

2. Write an ECP to display below mentioned patterns on 8-led's (4-active high led's and 4-active low led's)

   a. Switch ON one by one led from left to right at the rate of 1 second

   b. Switch ON one by one led from right to left at the rate of 1 second

3. Create a display on 8 LED's (4-active high and 4-active low) connected to port0 (0.0-0.7) in the given pattern.
   a. First all LEDs should be OFF, at 1st second, LED 0 and LED 7 (P0.0 and P0.7) should be ON
   b. At 2nd second, LED 1 and LED 6 (P0.1 and P0.6) should be ON
   c. At 3rd second, LED 2 and LED 5 (P0.2 and P0.5) should be ON
   d. At 4th second, LED 3 and LED 4 (P0.3 and P0.4) should be ON
   e. Repeat above pattern for 5 times and stop the process.
4. Write an ECP to display binary equivalent of 0-255 on 8-leds (4-active high led's and 4-active low led's)

Interfacing LED & SWITCH:

1. Create a display on 8 LED's (4-active high and 4-active low) using 4 active low switches connected to port0 (0.16-0.19) in the pattern explained:
   a. First all LEDs should be OFF.
   b) if 1st switch is pressed, LED 0 and LED 7 (P0.0 and P0.7) should be ON
   c) if 2nd switch is pressed, LED 1 and LED 6 (P0.1 and P0.6) should be ON
   d) if 3rd switch is pressed, LED 2 and LED 5 (P0.2 and P0.5) should be ON
   e) if 4th switch is pressed, LED 3 and LED 4 (P0.3 and P0.4) should be ON
2. Write an ECP to display binary equivalent of switch press count on 8-leds (4-active high leds and 4-active low leds)

3. PART1: Implement up and down counter. Use two switches and 8 leds (4-active high leds and 4-active low leds) as mentioned**:** sw1 for incrementing count, sw2 for decrement count and display updated count on leds.

Note: if count value is 0, at this time if sw2 pressed it should display 0 only on leds and if count value is 255 then if sw1 pressed it should display 255 only on leds.

Note: Time slice should start after the first switch press

4. Write an ECP to fulfill the below mentioned operations.

   1. Initially all leds should be off.

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2. Display even numbers on 8 leds with respect to 1 second if sw1 is pressed. Continue this process until sw2 is pressed.
3. Display odd numbers on 8 leds with respect to 1 second if sw2 is pressed. Continue this process until sw1 is pressed.

   Note: if number is >255 then roll over to 0 and repeat the operation.

Seven Segment Display + Switch:

1. Write an ECP to test seven segment display (all segments) is working or not.
2. Write an ECP to Show up counting from 0000 to 9999 on two multiplexed seven segments @ of 1sec (approximately).
3. Write an ECP to display the switch press count on two multiplexed seven segment display. (Note: if switch is pressed, need to display the updated count value after switch is released)

Interfacing 16X2 LCD:

1. Write an ECP to develop the driver for 16*2 alphanumeric LCD.
   a) To display a character
   b) To display a string
   c) To display an integer
   d) To display float number up to three decimal places.
   e) To display any custom character.
2. Write a program to display the message “VECTOR” on the first Line and “Institute” on the second line of a 2x16 LCD. Then make “Institute” flash at the rate of 1sec for 5 times, then clear the LCD screen
3. Write an ECP to rotate a string on LCD (From right to left)
4. Write an ECP display the basic time (HH:MM:SS) on LCD.

   Note: don’t use RTC registers
5. Write an ECP to display the switch press count on LCD.

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Assignments on 4x4 matrix keypad with output devices:

1. Write an ECP for displaying binary equivalent of a key Value from 4X4 matrix keypad on 8-LEDS (4-active high leds & 4-active low leds).
2. Write an ECP for displaying a key Value from 4X4 matrix keypad on 4-multiplexed seven segment displays.
3. Write an ECP to access a device (door/fan/bulb) with a valid pin/password entry from 4x4 matrix keypad.

Note: must include all test cases which are giving support for real-time applications

Assignments on UART:

1. Write an ECP to develop the driver for on-chip UART.
    a) To display a character
    b) To display a string
    c) To display an integer
    d) To display float number up to three decimal places.
    e) To display a hex value
2. Write an ECP for transmitting string through UART and display it on hyper terminal with 9600 baud rate.

    Example: “Welcome to VECTOR”
3. Write an ECP for receiving one character from PC through UART with 9600 baud rate and display its binary equivalent on 8 – LEDS (4 – Active high leds & 4 – Active low leds).
4. Write an ECP to infinitely receive any character then re-transmit its ASCII value in decimal and hex format. Output should be seen as for example below

Output: if Received character is ‘A’, then output should be 65 0x41

If Received character is ‘a’, then output should be 97 0x61

If Received character is ‘Z’, then output should be 90 0x5A and so on …

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